

Portable Sink Design

By: Chase Welliver, Colum Dougherty, Dylan Shoemaker

Element B Analysis

We researched water flow rate, bacteria levels, how to safely store water, how much water should be stored, plumbing, characteristics of children with learning/mobile disabilities, and even the benefits of children independently using sinks. With this knowledge, we analyzed several successful portable sink companies for how they balance these characteristics. While researching, we also noted the essentials of a sink we should prioritize while we design our own sink.

To start, the first sink we researched was made by a company called Boxio portable sinks. This sink is relatively small, and functions by utilizing a pressurized pump mechanism. The hand pump on the sink pressurizes the bottom clean water tank and allows the tube to siphon out the water and push it up out of the faucet. Something unique about this sink is that it is a lot smaller than most sinks on the market. This sink has the ability to collapse into smaller parts and also is easily transportable. The most notable strength of this sink that we may want to utilize ourselves is that it does not use battery or electric power, the ability to pump the sink up and have it run like a regular sink. The price of this sink is \$259 which is fairly reasonable for a portable sink. This sink was designed for campers who need a sink to clean a small amount of things or their hands. In our case, this sink wouldn't prove successful due to the fact that the Boxio sink does not accurately represent a regular sink with how it works. The sink also does not fit the full parameters that are needed to accurately fit our requirements. One main requirement that is not met here is the adequate flow rate. The sink works off of a hand pump and each pump only allows a certain amount of time in which the sink can run. The sink can only run for a small amount of time and can be quite laborious to constantly have to pump the sink to run long enough to wash your hands to a degree that is acceptable.

Another sink we looked at were Monsam Enterprises Portable Sinks. This company offered pretty generic sink designs with many being made at a low height for children. One sink we analyzed was 25 inches off the ground, which seems like a good height for a child. The sink at Whitemarsh was 24 inches tall, which further supports this decision. Something to consider for our sink is that it will be picked up and is able to be placed on a student's desk. However, the sink (which will also have the water storage below it) may be too tall if placed on a desk. A common theme we noticed is that many of these sinks have wheels on the bottoms to facilitate portability. When we were interviewing whitemarsh, we did not originally consider this; but rather assumed our sink would be lightweight enough to pick up. However, if there is no way around reducing the weight of the sink, this may be our only solution. When it comes to

strengths and weaknesses of this sink, this sink seems to sacrifice portability and small size for a more appealing design, storage, and utilities. The typical price of their sinks is roughly 2,000 dollars. This is mainly due to the high quality of the materials and design. This sink would be acceptable in the classroom but not ideal. This sink has limited portability which we decided is a top priority. Overall, although this sink has some exceptional features, there are many modifications we would want to make to better suit W.E.S. students.

Another notable sink we researched is Washing Hands' Portable Sinks. This company probably provided the highest quality of sinks yet. In addition to the previously mentioned sink, it was clear this sink was not meant to be moved around often, especially via being picked up. Another important note is that many of the researched sinks provided water heating and cooling. This is a substantial addition to the complexity of the project, which is not required in our case. Overall, while this is a visually appealing and sturdy sink, it is definitely too large and bulky for our standards. Although this sink seems suitable for maybe a restaurant, it is not ideal for our classroom where we require a shorter and lighter sink. The foot pedal also does not match our needs.

Overall, we've noticed that in the past, the main material of successful sinks often involved metal, wood, and occasionally plastic. We believe that in terms of difficulty to build with these items, wood would be the easiest, and plastic and metal would provide the most challenges. However, wood is prone to water damage, decay, mold, and other hygienic, durability, and cleanliness issues. Between plastic and metal, our group agreed that plastic is more appropriate for our goals. Its main strength is its lightweight portability. Other notable successful criteria includes the impermeable structure of plastic and low price. Our primary concern is how we could shape plastic to suit our goals. One possible solution we have discussed is buying a plastic box already shaped to our specifications.

[How to Calculate the Home Water Flow Rate](#)

- Flow Rate= Volume of Fluid(Liters or gallons)/Time(seconds)
- Flow rate is an important factor when determining specific plumbing plans within specific homes.
- Flow rate can help diagnose plumbing issues that are affecting sewage systems.
- To ensure the effectiveness of this formula, it is essential to gauge the quantity of fluid traversing a pipe within a designated time frame. Follow these instructions:
 1. Quantify the water volume: Utilize a container with a predetermined volume, measuring the duration it takes to reach capacity.
 2. Time the flow: Initiate a timer as the fluid begins filling the container and cease it upon reaching full capacity.
 3. Calculate the flow rate: Divide the fluid volume by the elapsed time (V/t). This computation yields the flow rate, expressed in gallons per minute or liters per minute.

[Plumbing Basics | HowStuffWorks](#)

- Majority of drainage systems utilize the force of gravity to allow the waste to flow through pipes and to eventual sewage treatment facilities
- Traps, air vents, and clean outs are key to a functional drainage system with minimal issues.
- Proper slope for pipes that are carrying waste horizontally is a ¼ inch vertical drop per foot of pipe used.
- Air supply to the pipes is key to allow the waste to flow through the pipes properly
- Supply and Drainage systems must be segregated from each other. Otherwise, there is potential for sanitary issues.

[Safe Water Storage | Global Water, Sanitation and Hygiene | Healthy Water | CDC](#)

- Water must be stored at a temperature that does not cause the water to freeze or boil.
 - Durable water storage is made out of hard-plastics, metals, and ceramics.
 - Water storage systems recommend a standard volume of water of 20 liters
- Unscented chlorine bleach is a recommended sanitation agent for water storage systems
- Water storage systems should not be in the presence of sunlight as it can alter the storage material and cause potential risks for users.

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[What conditions encourage bacteria to grow? / Food Safety FAQ / Health and Safety / Consumer Resources / Home](#)

- Bacteria can live in a variety of temperatures
- Bacteria thrives in warm moist environments
- Environments with neutral or slightly acidic often allow bacteria to thrive

